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Batch : C3 / DV (Fri)

TITLE: Introduction to Tableau

Objective:

To introduce the fundamentals of Tableau and enable students to create basic interactive visualizations using data.

Expected outcome of Experiment:

CO1	**Learn how to locate and download datasets, extract insights from that data and present their findings in a variety of different formats.**

Books/ Journals/ Websites referred:

https://www.tableau.com/

Background Theory:

Tableau offers an easy-to-use, self-service analytics cloud platform designed to deliver insights. With an at-a-glance visual representation of the data and behind-the-scenes automation, Tableau automatically enriches analytics data with business context and meaning helping to discover and understand relevant data.

Laboratory Work:

Refer this link: https://help.tableau.com/current/guides/get-started-tutorial/en-us/get-started-tutorial-connect.htm

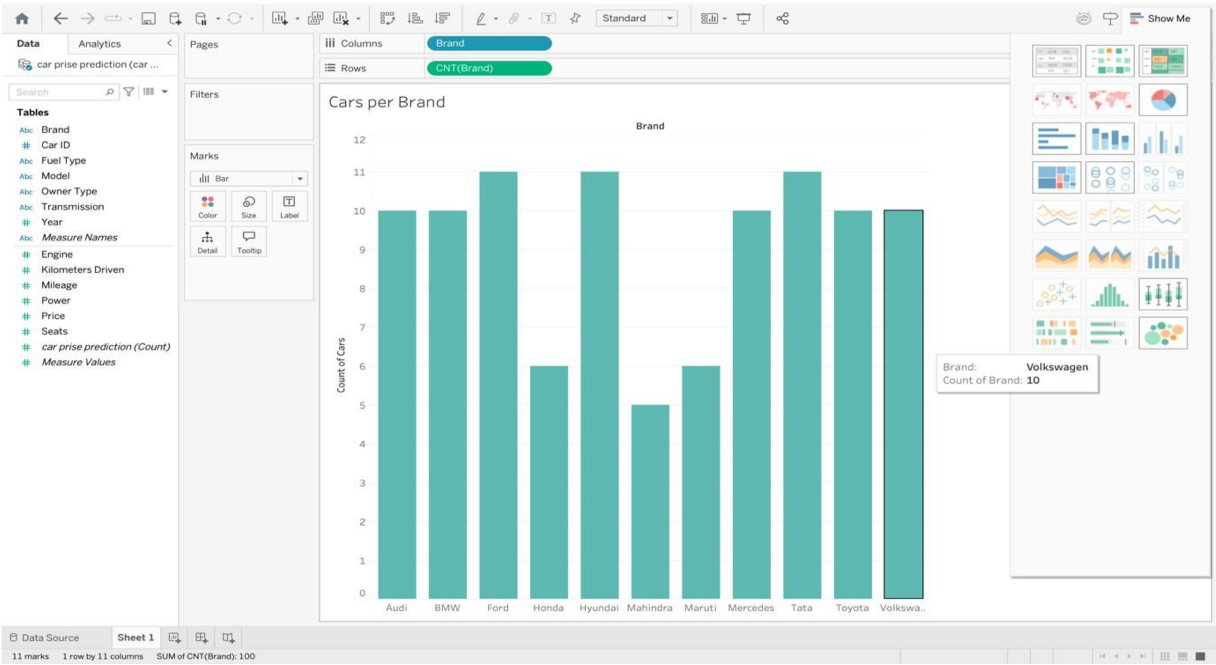
Perform these tasks

1. **Connecting to Data Source and Visualizing one specific table**
2. **Creating and Refining a view**
3. **Focus results**
4. **Exploring data geographically**
5. **Drill down into the details**

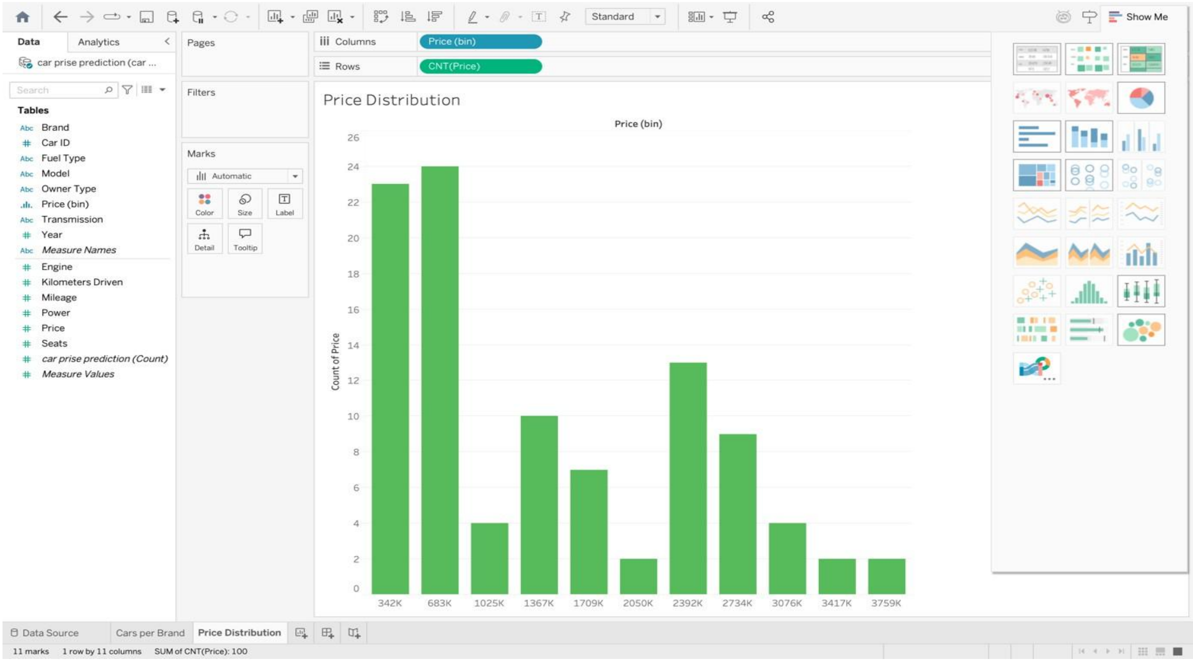
Practice these tasks for Superstore (default dataset available in Tableau) and one more dataset like [HR analytics dataset]() or [Customer Profiling dataset]()

Implementation screenshots:

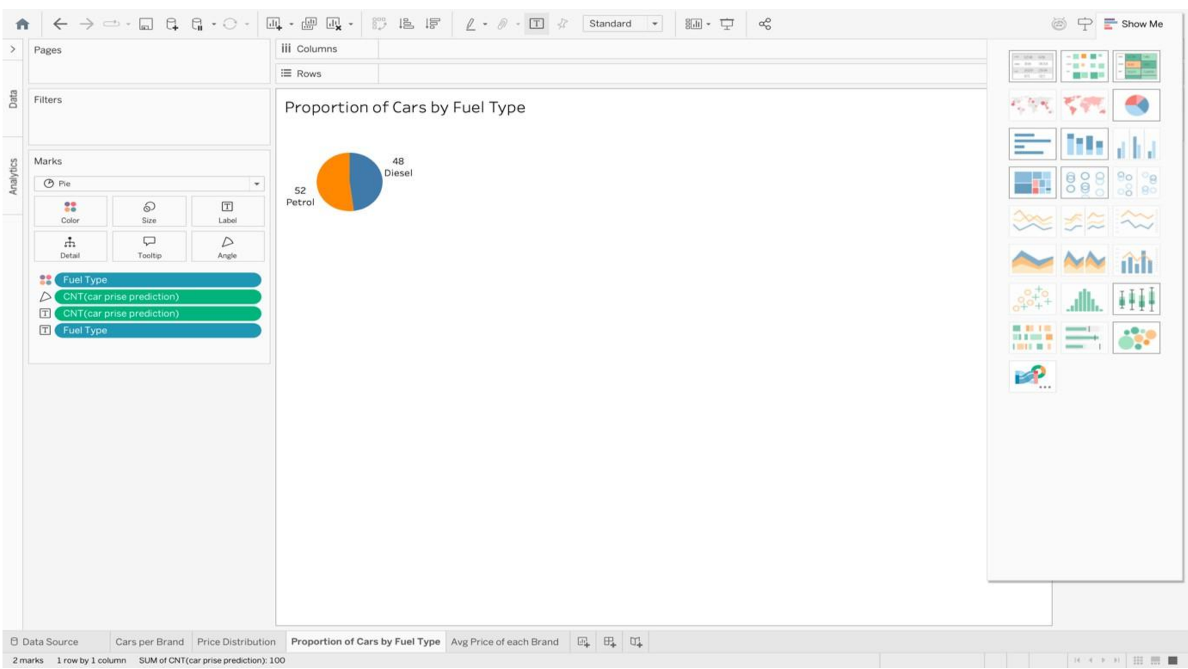
1) Bar chart showing the number of cars available for each brand



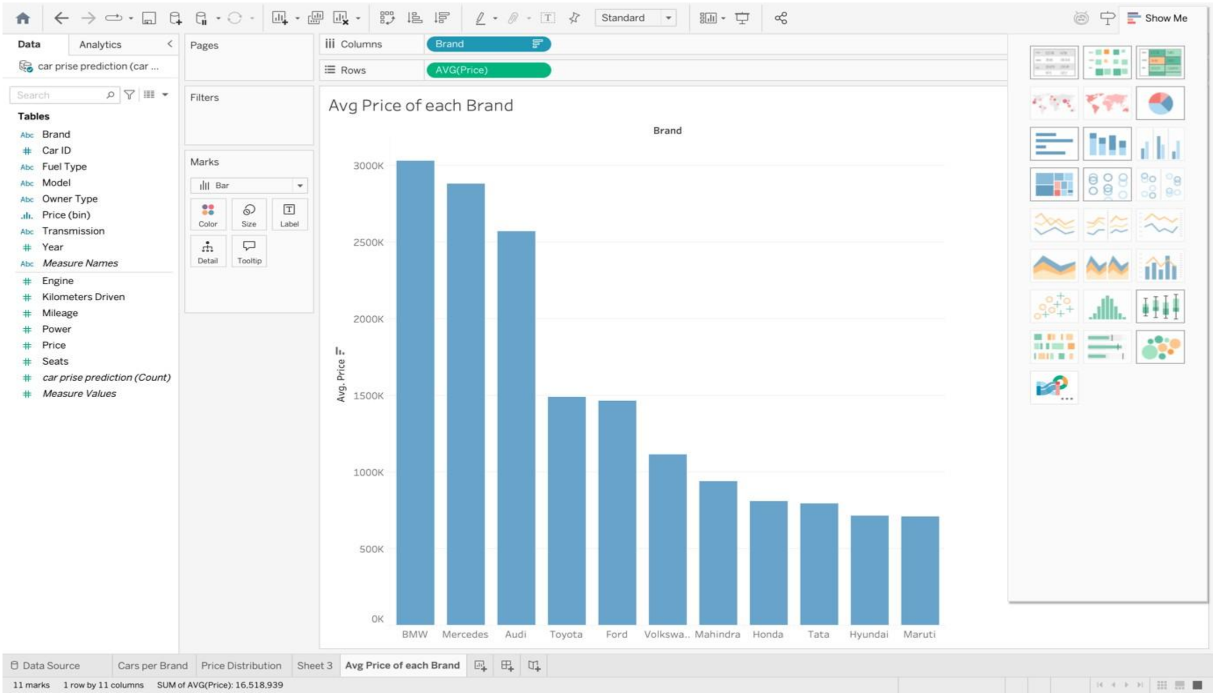
2) Histogram of car prices to see the overall distribution



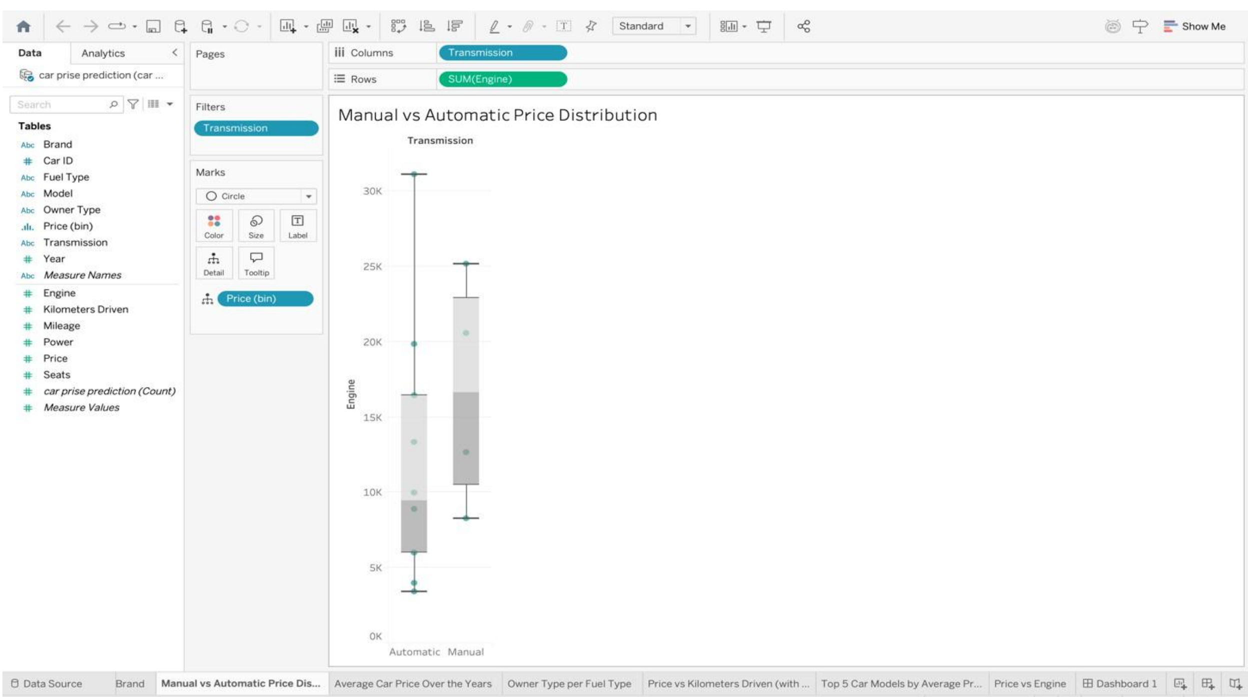
3) Pie chart showing the proportion of cars by fuel type



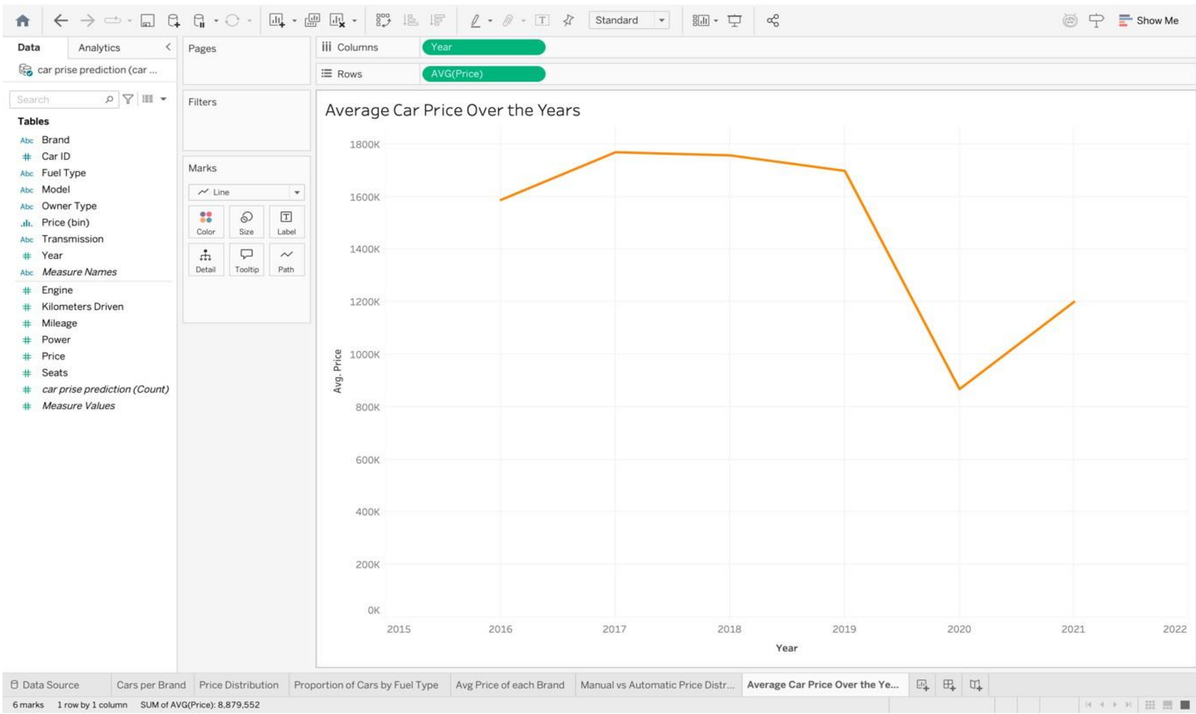
4) Bar chart showing the average price of cars for each brand



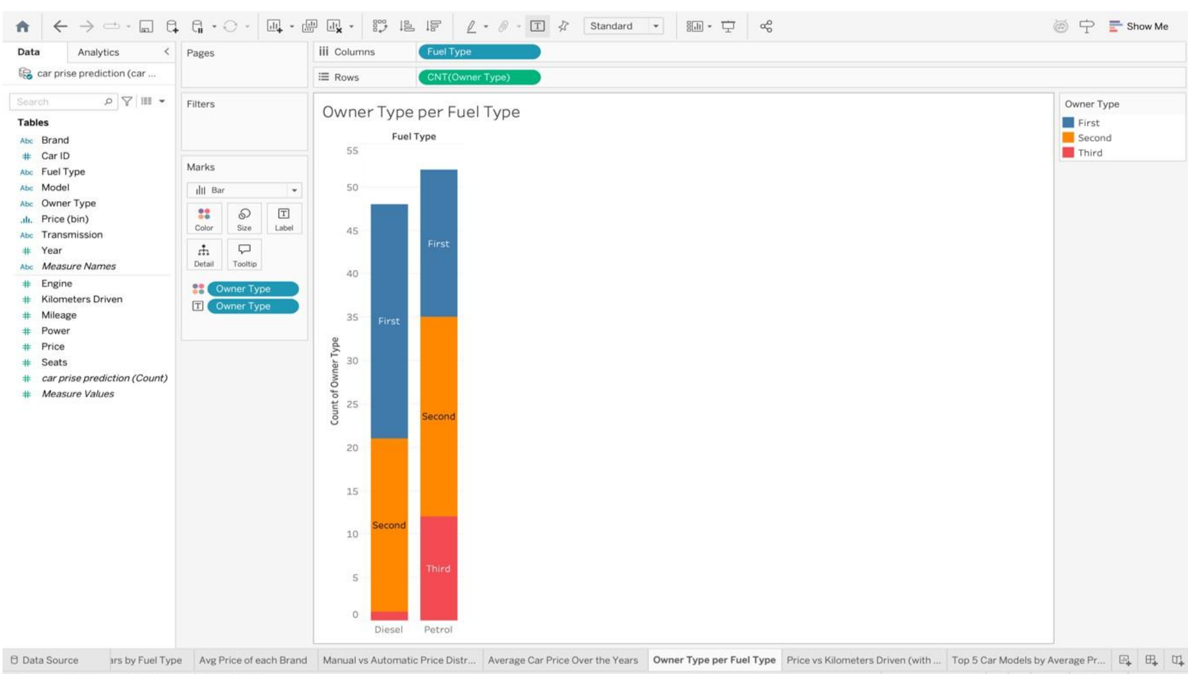
5) Box plot comparing the price distribution for Manual vs Automatic cars



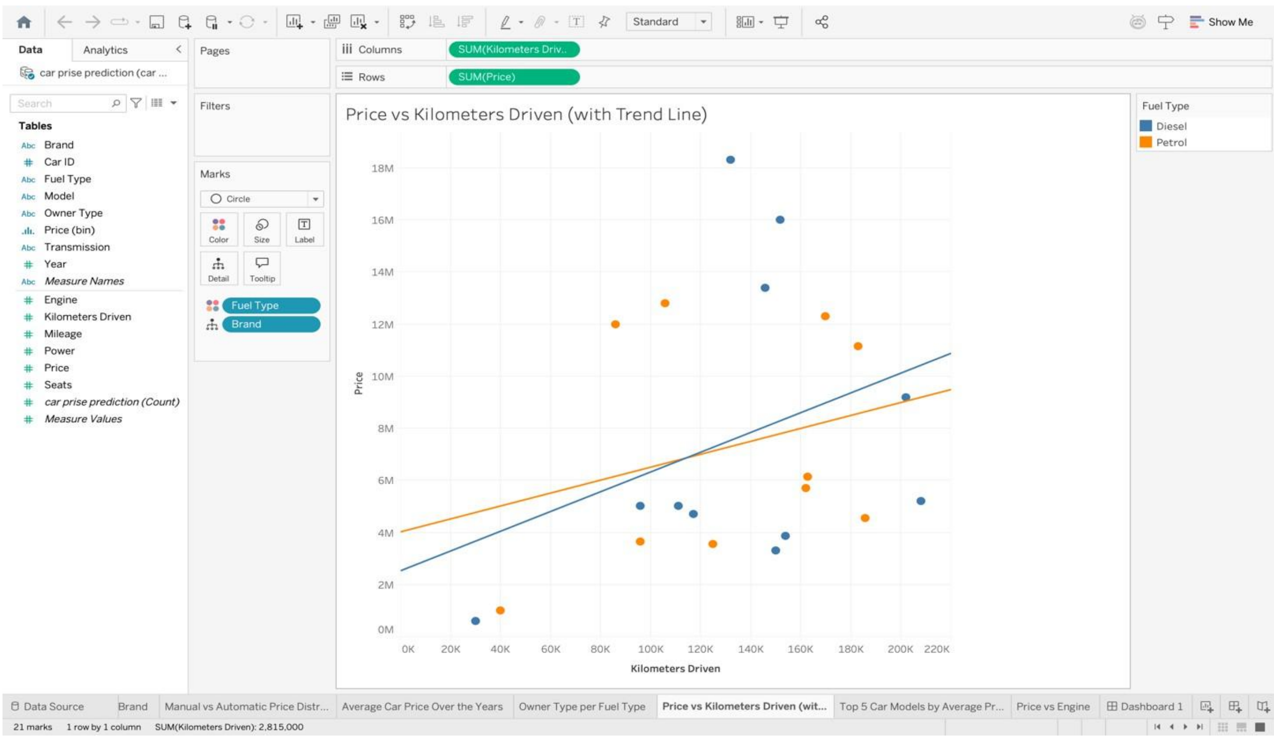
6) Line chart showing average car price over the years



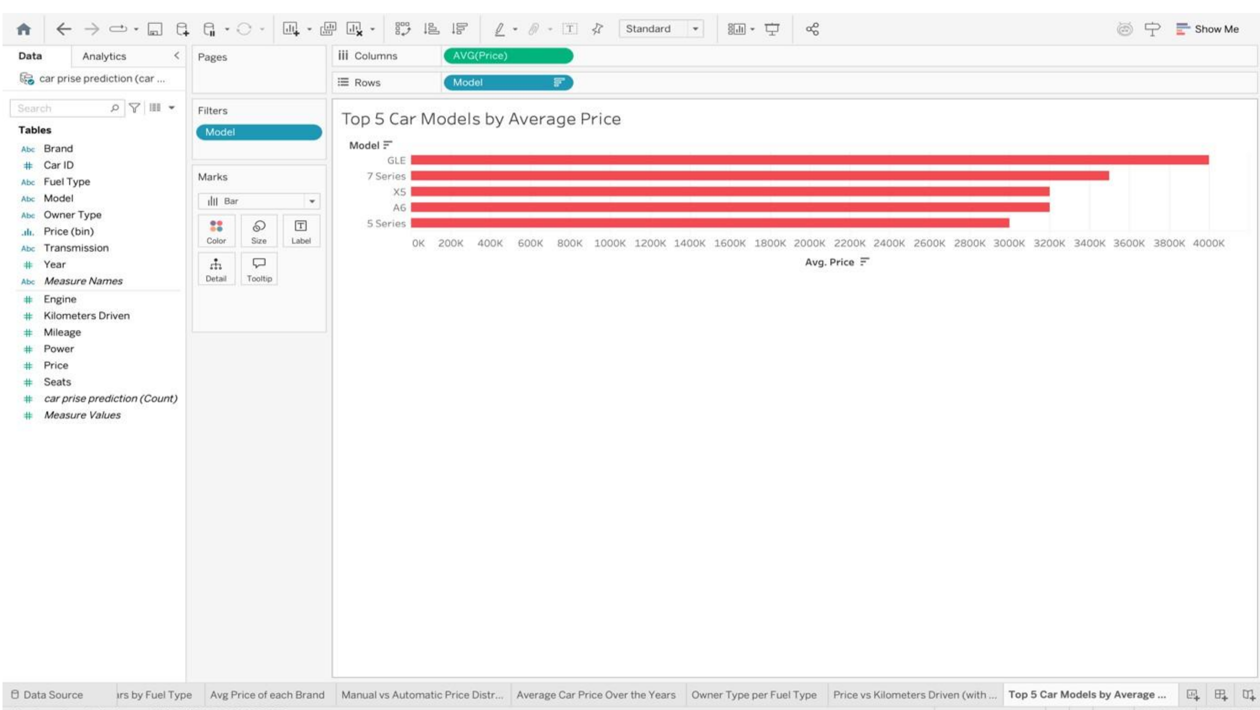
8) Stacked bar chart showing the count of cars in each owner type per fuel type



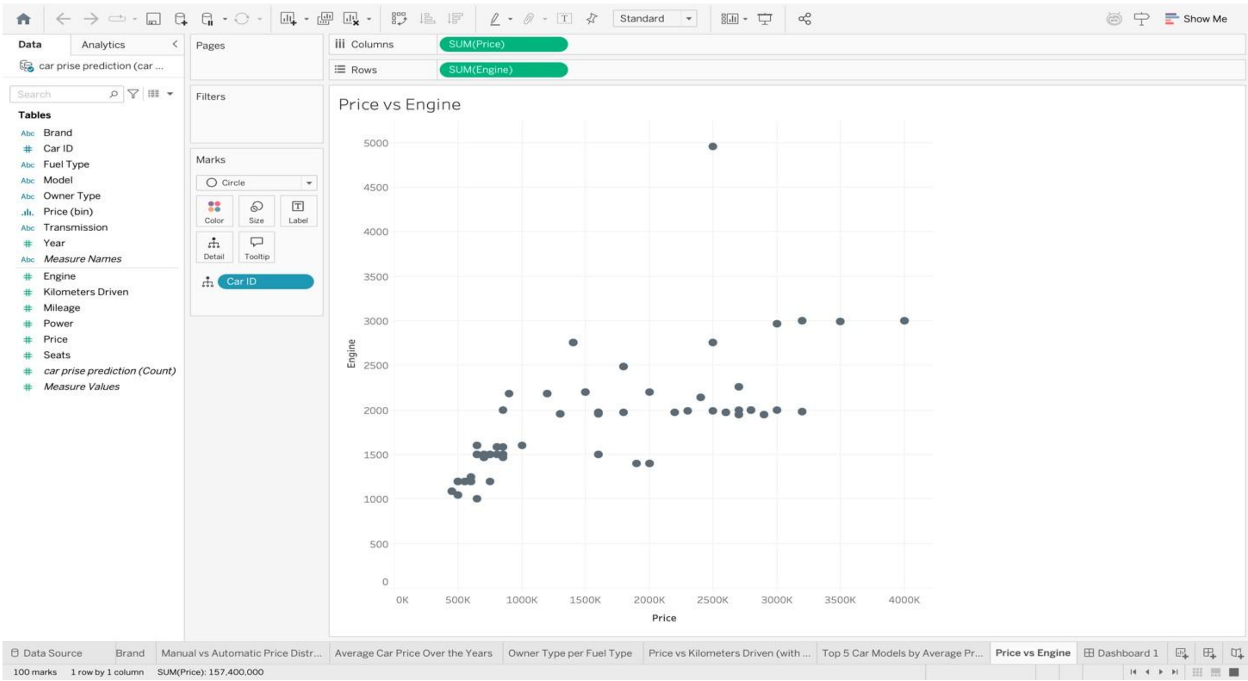
9) Scatter plot of car price vs kilometers driven, colored by fuel type, and added trend line.



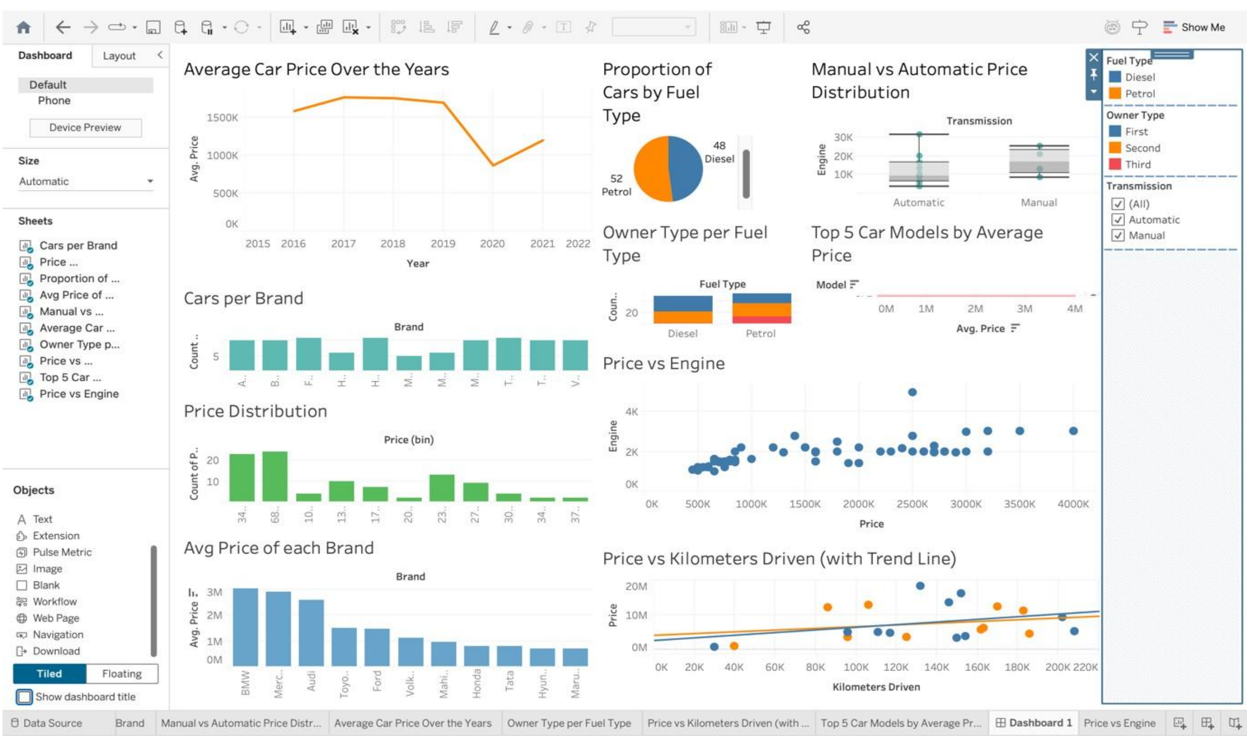
10) Show a bar chart of the top 5 car models with the highest average price



11) Scatter plot of price vs engine size



12) Dashboard showing price by brand, with filters for fuel type, transmission, and owner type



Conclusion:
The above experiment gives an introduction into various features of tableau, use of in-built measures to create different analytical visuals, the varied application of each chart and finally building a comprehensive dashboard with filters and anlaysis.

Post Laboratory Questions

1. **What do you mean by refining a view in Tableau? List any two refinement actions performed during the experiment**

   Refining a view in Tableau means modifying and improving a visualization to make the data clearer, more meaningful, and easier to analyze. This is done by adjusting filters, sorting, formatting, or adding details so that insights can be observed more effectively.

   Two refinement actions:

   1. **Applying filters** to display only relevant data (e.g., filtering by car brand or fuel type).
   2. **Sorting data** in ascending or descending order to compare values easily.

2. **Why is it important to select and visualize a single specific table instead of the entire dataset while starting analysis?**

   Selecting a **single specific table** helps in:

   - Reducing **complexity and confusion** during initial analysis.
   - Improving **performance and faster loading** of visualizations.
   - Allowing **focused and accurate insights** by analyzing only relevant data.
   - Avoiding incorrect conclusions caused by **unnecessary or unrelated fields**.

   This makes the analysis more structured and easier to interpret.

3. **Which graph type would you choose to:**
   a) Compare values across categories
   b) Show trends over time
   c) Explore geographical data
   Justify your choices.

   a) Compare values across categories

   Bar Chart
   Justification: Bar charts clearly show differences between categories such as brands, fuel types, or models.

b) Show trends over time

Line Chart
Justification: Line charts are best for visualizing changes and trends over a period of time, such as price variations across years.

c) Explore geographical data

Map (Geographical Chart)
Justification: Maps display data based on location, making it easy to analyze regional or location-based patterns.